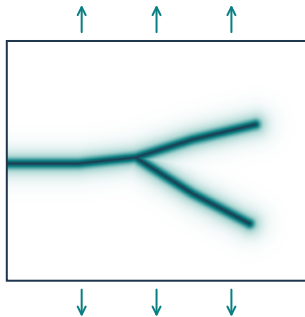


From cracks to computation

Phase-field fracture, differentiable
FEM and learning with PhAST

Three sessions. Six hours.
A shared mechanics example.



Original schematic, not a simulation

One specimen. Three learning questions.

A Why does a crack grow?

B How do we compute and differentiate it?

C When can a learned proposal be useful?

Six-hour course design: 330 minutes of teaching/activity + 30 minutes of breaks. Not a claim about the published event timetable.

Understand the crack

Session A / 00–10

Read a damage field.

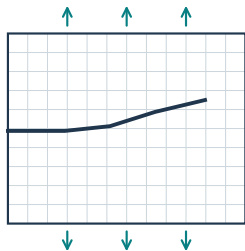
Explain the energy.

Trace one numerical increment.

120 minutes, including a 10-minute break

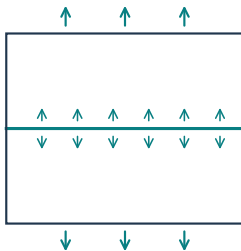
Three descriptions of an evolving crack

Sharp discontinuity



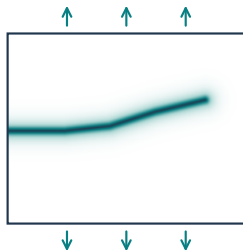
XFEM enriches the approximation around a sharp discontinuity.

Cohesive interface



A cohesive law relates interface traction to separation.

Diffuse phase field



Phase field adds damage d and length scale ℓ .

Original schematics; these approaches can be combined. Miehe, Welschinger & Hofacker (2010), [IJNME 83, 1273–1311](#).

A model is not a solution algorithm

Question	Examples
What does the crack mean?	Sharp surface, cohesive relation, diffuse field
How is it discretised?	FE spaces, enrichment, mesh, quadrature
How is it solved?	Staggered/monolithic; Newton/quasi-Newton

Yes: quasi-Newton is a nonlinear solution method, not a rival crack model.

Fracture balances storage and crack cost

A representative small-strain brittle model

$$\mathcal{E}_\ell(u, d) = \underbrace{\int_{\Omega} \left[g(d) \psi^+(\varepsilon(u)) + \psi^-(\varepsilon(u)) \right] dx}_{\text{elastic energy: damage reduces tensile storage}} + \underbrace{\frac{G_c}{c_0} \int_{\Omega} \left[\frac{w(d)}{\ell} + \ell |\nabla d|^2 \right] dx}_{\text{crack cost: local damage + spatial regularisation}} - \underbrace{W_{\text{ext}}}_{\text{loading}}.$$

Francfort & Marigo (1998), [JMPS 46, 1319–1342](#). Bourdin, Francfort & Marigo (2000), [JMPS 48, 797–826](#). Split shown is representative; dynamic mechanics also requires inertia.

AT1 and AT2 choose the crack-density term

$$c_0 = 4 \int_0^1 \sqrt{w(s)} \, ds$$

AT1

$$w(d) = d, \quad c_0 = \frac{8}{3}$$

Finite elastic stage in the standard idealisation.

$$\underbrace{d=0}_{\text{intact}}$$

$$\underbrace{d=1}_{\text{broken}}$$

AT2

$$w(d) = d^2, \quad c_0 = 2$$

Gradual early damage in the standard idealisation.

$$d_{n+1} \geq d_n$$

Degradation $g(d)$ is a separate choice. Pham et al. (2011), [IJDM 20, 618–652](#).

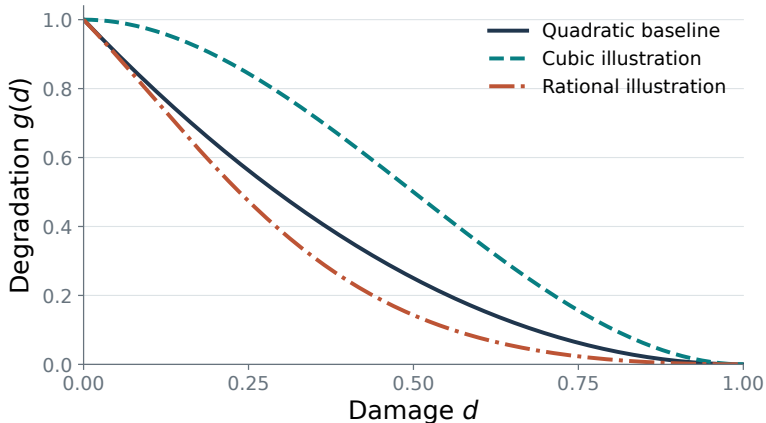
How quickly should stiffness be lost?

Quadratic baseline

$$g(d) = (1 - \eta)(1 - d)^2 + \eta$$

$$g(0) = 1$$

$$g(1) = \eta$$

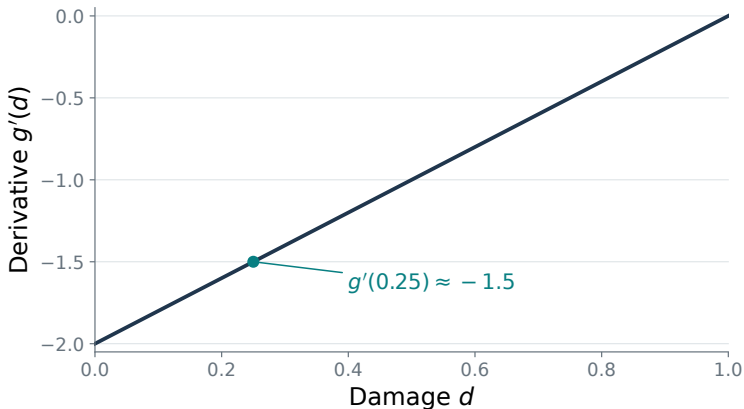


Analytic illustrations, $\eta = 10^{-6}$; not calibrated PF-CZM. Notebook 01/02 uses $\eta = 10^{-7}$.

A derivative measures the local energy response

$$g'(d) = -2(1 - \eta) \cdot (1 - d)$$

$$g''(d) = 2(1 - \eta)$$



Local analytic derivative; not a full fracture-path derivative. 10-minute break follows this discussion.

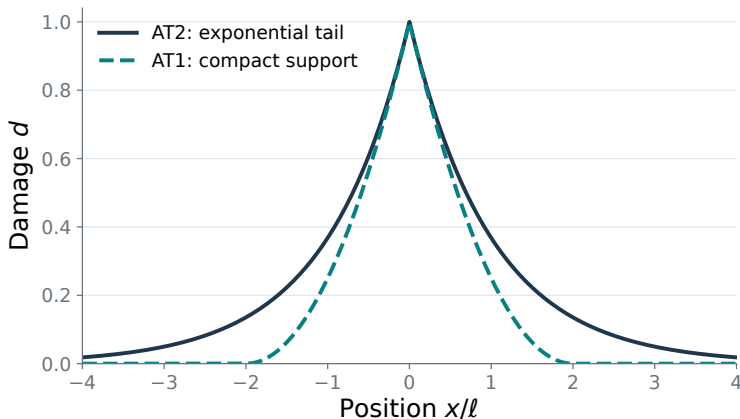
A diffuse crack has a spatial profile

AT2

$$d = e^{-|x|/\ell}$$

AT1

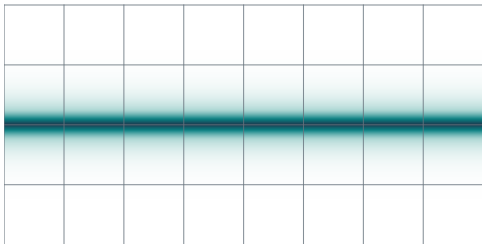
$$d = \left[1 - \frac{|x|}{2\ell}\right]_+^2$$



Isolated 1D crack-density minimisers, $d(0) = 1$; no mechanics solve. Pham et al. (2011), [IJDM 20, 618–652](#).

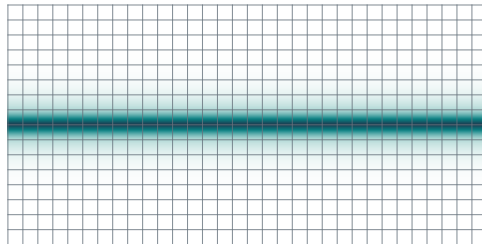
The mesh must resolve the chosen length

Coarse sampling



ℓ : model scale

Finer sampling

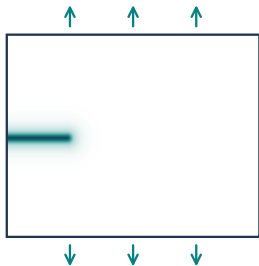


h : discretisation scale

Same imposed analytic band; two schematic meshes. No universal h/ℓ threshold or convergence claim is implied.

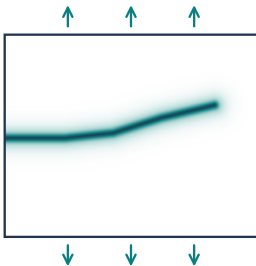
Initiation is not propagation or branching

Initiation



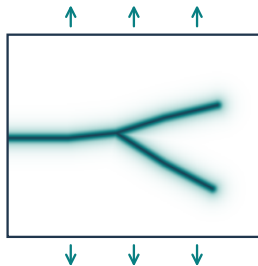
Where does damage first grow?

Propagation



Where does an existing crack advance?

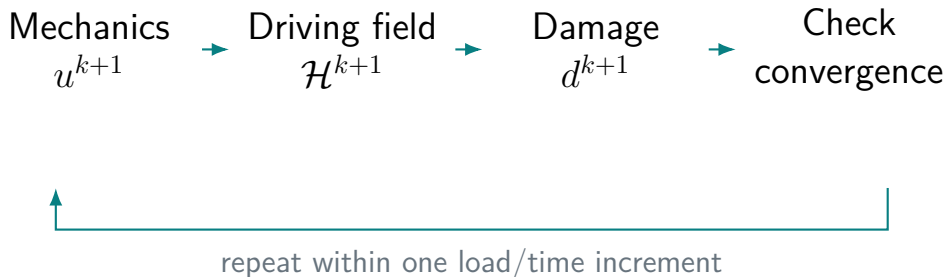
Branching



When does one path become two?

Original conceptual states, not a simulation sequence. Branching claims require whole-field, load/time and numerical evidence.

A staggered step alternates coupled problems



History-field notation is representative; irreversibility and the convergence criterion depend on the selected formulation. Miehe, Welschinger & Hofacker (2010), [IJNME 83, 1273–1311](#).

Time integration and nonlinear iteration are different

$$\underbrace{\mathbf{M}\ddot{\mathbf{u}}}_{\text{inertia}} + \mathbf{r}_{\text{int}}(\mathbf{u}, d) = \mathbf{f}_{\text{ext}}$$

Static/quasi-static	Equilibrium; a load path need not be physical time.
Explicit mechanics	Central difference/Verlet; stability limits the time step.
Implicit mechanics	Newmark/generalised- α ; solve an implicit update.
Newton/quasi-Newton	Iterations for a nonlinear system, not a time discretisation.

An explicit mechanics update can still use an implicit damage solve. Notebook 01 follows a quasi-static load history.

Matrix-free means applying an operator

$$\mathbf{y} = \mathbf{K}\mathbf{v} \quad \longleftrightarrow \quad \mathbf{y} = \sum_e \mathbf{A}_e^T \mathbf{K}_e \mathbf{A}_e \mathbf{v}$$



Conceptual linearised FE operator. Matrix-free avoids storing an assembled global matrix, not all storage or all solver cost. [PhAST source and documentation](#).

Tensors still have physical locations

Field	Typical shape	Meaning
Coordinates	$N \times n_{\text{dim}}$	positions
Connectivity	$N_e \times n_{\text{loc}}$	local-to-global map
Displacement, damage	$N \times n_{\text{dim}}; N$	nodal fields
Strain, history	element/quadrature axes	local constitutive state

Illustrative layouts, not an exact API specification. The executable notebook/source determine ordering.

Build, run and differentiate

Session B / 00-04

Specify a problem.

Read its fields and response.

Check a derivative before recovery.

120 minutes, including a 10-minute break

Start from the same computational state

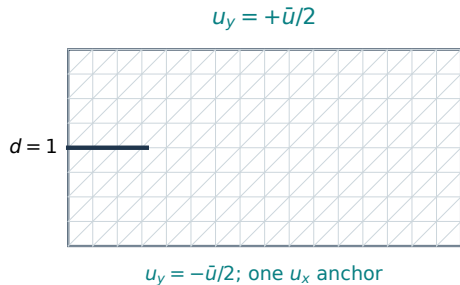
Notebook 01 / setup and version record



PhAST 0.16.2 f6324f899f07

Use the exact bundled notebook setup, not a generic install command. Fresh Colab timings remain unverified. [PhAST source and documentation](#).

Geometry, mesh and precrack are separate inputs



Fast route: rectangular T3 mesh from tensors.

The precrack is prescribed $d = 1$ on centreline nodes.

Geometry/import extension: create a Gmsh mesh and inspect named groups.

Original mesh/BC schematic: symmetric opening and a prescribed damage precrack.

A named boundary is not yet a boundary condition



$$\text{top} \longrightarrow \{i_1, i_2, \dots\} \longrightarrow u_y \longrightarrow \bar{u}$$

Inspect group membership and coordinates before solving. The quick tensor-mesh route identifies boundary node sets; imported meshes require their own group checks.

A reproducible calculation has a complete specification

Input	Tiny public PhAST route
Material / fracture	E, ν, G_c, ℓ, η ; AT2, isotropic degradation
Mesh / precision	T3; 128×64 cells; float64
Loading	60 quasi-static increments; symmetric opening
Numerics	SciPy mechanics; damage/outer tolerances; failure checks

Exact values and units/convention belong to `configs/day2_forward/tiny_notched_tension.json`. No dynamic time or matrix-free benchmark is claimed.

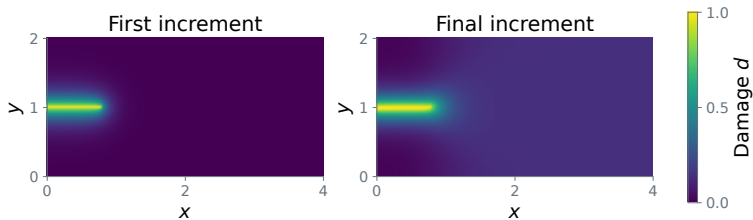
Distinguish the seeded crack from new damage

AT2 crack cost

$$\gamma_{\ell}(d) = \frac{d^2}{2\ell} + \frac{\ell}{2} |\nabla d|^2$$

$$G_c \int_{\Omega} \gamma_{\ell}(d) \, dx$$

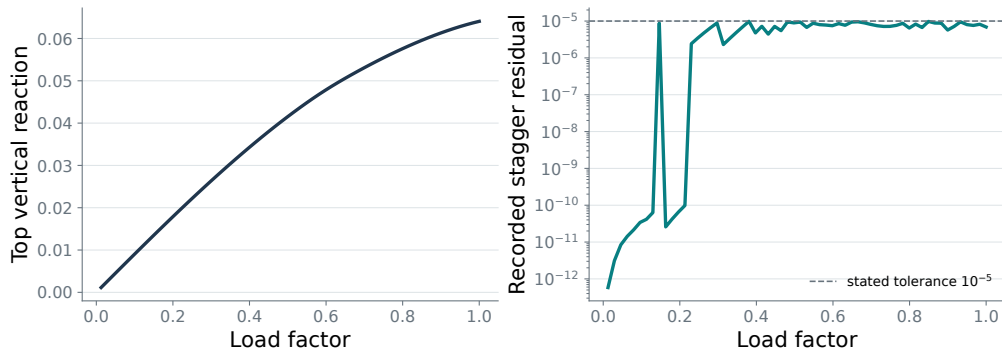
Seed: prescribed $d = 1$



History: $d_{n+1} \geq d_n$

Actual PhAST damage evolution, 60 quasi-static increments; source f6324f899f07. Not branching.

Read response and numerical checks together



Actual notebook 01 trace. Residual convergence is not mesh convergence or full physical validation.

Change one input, then explain the difference

Five-minute comparison / followed by a ten-minute break



A material/load change or a numerical refinement

Changing ℓ is not simply mesh refinement. Label saved outputs and live results separately.

Tensor code can mirror the finite-element calculation

```
u_e = u[connectivity]
strain_q = B @ u_e
stress_q = law(strain_q, d_q)
r_e = integrate(B.T @ stress_q)
r = accumulate(r_e, connectivity)
```

Gather

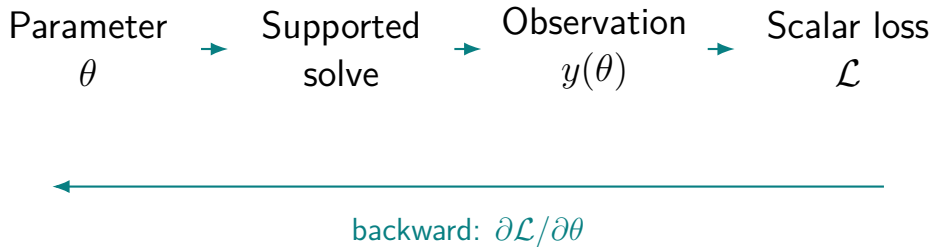
Local physics

Weighted quadrature

Global accumulation

Illustrative pseudocode, not an executable PhAST API. Actual implementation may use batched/custom operators.

A gradient follows a specific computational path



Custom solves, convergence, boundary treatment, detach, clamps, history maxima and active sets matter. [PyTorch autograd documentation](#).

Check a local derivative three ways

Notebook 02 / analytic, automatic, finite difference

$$g'(d) = -2(1 - \eta)(1 - d) \quad D_{\epsilon}g(d) = \frac{g(d + \epsilon) - g(d - \epsilon)}{2\epsilon}$$

```
d = torch.tensor(0.25, dtype=torch.float64,  
                 requires_grad=True)  
(g_ad,) = torch.autograd.grad(material.degradation(d), d)
```

Exact PhAST local law: $\eta = 10^{-7}$, $g'(0.25) = -1.49999985$. This is not a full fracture-path derivative test.

Choose an informative inverse observation

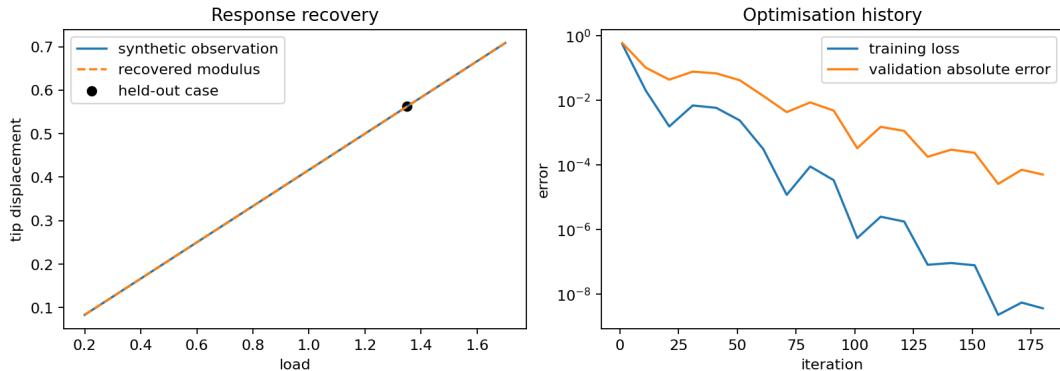
Notebook 03 / original elastic-bar toy, not fracture inversion

$$u_{\text{tip}}(E) = \frac{FL}{EA} \quad \mathcal{L}(E) = \frac{1}{2} \sum_j \left[u_{\text{tip}}(E; F_j) - u_j^{\text{obs}} \right]^2$$



Fixed-mesh, one-parameter teaching problem. Optimising $\log E$ preserves positivity; it does not establish identifiability in a more complex model.

A small loss is only one part of recovery evidence



Actual executed notebook 03 plot: original elastic-bar toy. Not a PhAST fracture inverse result or an unpublished paper exhibit.

Keep an experiment someone else can reproduce

1. Configuration and source version.
2. Whole field, response and numerical checks.
3. Derivative check and inverse interpretation.

Session B hand-in: one executed notebook and one comparison sheet, including precision, hardware and measured timing scope.

Train, save and assess a model

Session C / 00-04

Fit a compact model.

Reload the same prediction.

Assess a proposal and retain correction.

120 minutes, including a 10-minute break

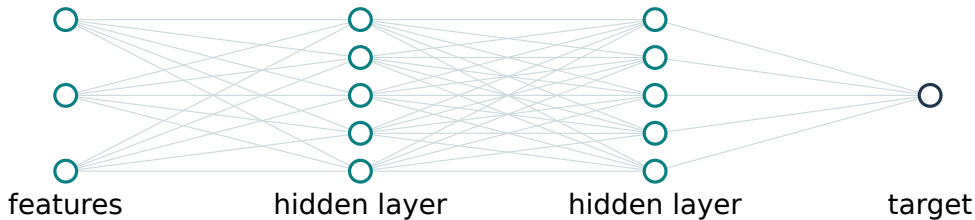
The data card defines what learning means

Notebook 04 / course-owned field equation

Item	Teaching example
Input	Nodewise $[x/L, y/H, \text{load factor}]$
Target	A scalar damage-like toy field, not a fracture field
Split	Whole load cases: train, validation, held-out test
Provenance	Discrete Helmholtz-like reference equation

Separate whole cases/trajectories to prevent leakage. Feature order, units, location and normalisation are part of the model contract.

An MLP maps named features to a named target



$$\hat{d} = \mathcal{N}_{\omega}(\text{normalise}(x/L, y/H, \lambda))$$

Original architecture schematic, not the exact layer count. A pointwise model does not acquire mesh connectivity merely from coordinates.

A fitted model can still miss the local field

Toy reference

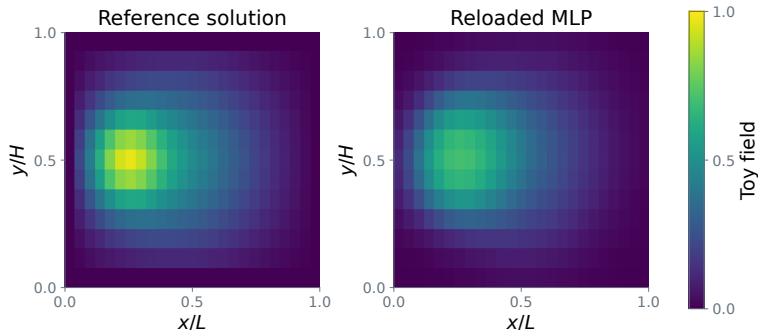
$$(\mathbf{I} + \ell^2 \mathbf{L}_h) d = b$$

Model

$$\hat{d} = \mathcal{N}_\omega(x/L, y/H, \lambda)$$

Same held-out load.

Same colour scale.



Executed notebook 04: original Helmholtz-like teaching problem, not fracture data. Held-out load $\lambda = 1$.

A checkpoint is more than weights

Notebook 04 / save, new instance, reload, compare

```
checkpoint = {  
    "state_dict": model.state_dict(),  
    "architecture": configuration,  
    "features": feature_order,  
    "normalisation": scaling,  
    "source": data_provenance,  
}
```

$$\max_i \left| \hat{d}_i^{\text{before}} - \hat{d}_i^{\text{reload}} \right| \leq \tau$$

Illustrative checkpoint layout; use the notebook's exact schema. Preserve shape, precision, evaluation mode and supported scope. [PyTorch autograd documentation](#).

A model swap requires compatible inputs

Representation	Required input construction
MLP / RBF	Named point features; scaling; RBF centres/kernel
GNN / GNO	Nodes, edges/neighbourhoods and geometry/operator data
CNN	Grid/channel order and explicit mesh-grid maps

The notebook MLP/RBF share a narrow toy contract. An architecture name does not guarantee mesh transfer. Followed by a 10-minute comfort break.

A learned model supplies a proposal

Notebook 05 / a defined teaching adapter



Detached inference $\neq$ training through the solver

Notebook 05 uses a toy residual and no-gradient prediction. It is not an integrated PhAST AT2 surrogate or a training-through-fracture demonstration.

Bounds are necessary but not equilibrium

$$d^{\text{proj}} = \max\left(d^{\text{previous}}, \min(1, \max(0, \hat{d}))\right)$$

Projection can enforce

$$0 \leq d \leq 1$$

$$d \geq d^{\text{previous}}$$

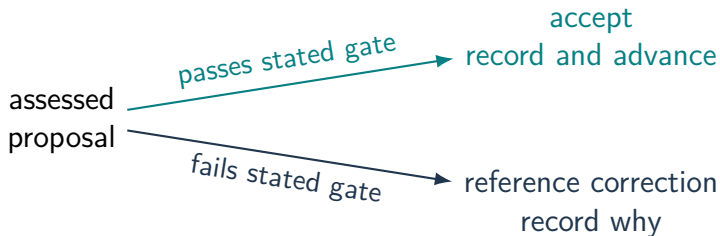
Assessment still asks

Are the equations satisfied?

Are prescribed values consistent?

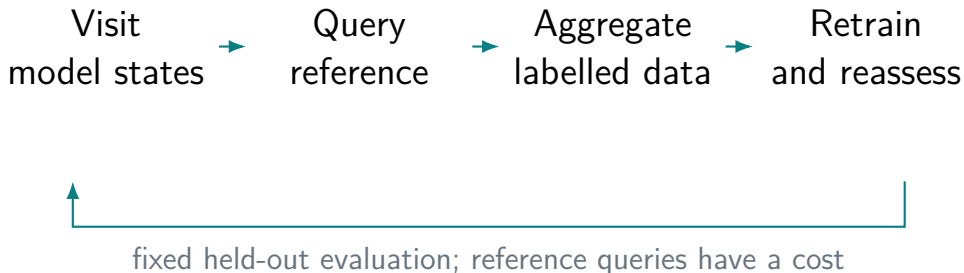
Use a residual/projected-KKT criterion appropriate to the actual route. Projection alone does not establish energy consistency or a plausible crack path.

Acceptance needs an explicit correction route



Log proposal, projection, residual, decision, correction and total cost. A successful toy gate is not a universal reliability or acceleration claim.

DAGger learns from states the model visits



Ross, Gordon & Bagnell (2011), [AISTATS: dataset aggregation](#). Notebook 05 logs a replay record; a full aggregated retraining round is optional, not claimed as executed.

Architectures encode different assumptions

Point map	Compact features; explicit coordinate assumptions
Grid or graph	Structured stencil or irregular neighbourhood
Operator map	Function inputs/outputs; training and discretisation scope

MLP/RBF, CNN, GNN/GNO and neural-operator names describe representations, not a ranking or automatic generalisation guarantee.

Read a research example through the same evidence

1. Whole field and physical load/time history.
2. Matched response, numerical checks and held-out scope.
3. Source/configuration and complete computational cost.

Discussion framework, not a substituted or invented paper result. Add a permitted research exhibit only with its actual provenance and limitations.

A useful contribution starts with reproducibility



github.com/CEMS-Lab/PhAST

Read current contribution and issue guidance before submitting. Keep private data, credentials and unpublished results out of public reports.

Keep the chain from assumptions to evidence visible

A Representation $\rightarrow$ energy $\rightarrow$ discretisation

B Configuration $\rightarrow$ result $\rightarrow$ derivative

C Data $\rightarrow$ prediction $\rightarrow$ assessment

Full references and course links: [slides/README.md](#) and the companion book. Original graphics; no CWI slides were copied.